Centralized and Federated Learning for COVID-19 Detection With Chest X-Ray Images: Implementations and Analysis

Sadaf Naz, Khoa Phan, and Yi-Ping Phoebe Chen

Abstract—**In the health domain, due to privacy issues, many important datasets are isolated, which nonetheless need to be analyzed collaboratively for conclusions to be drawn efficiently. To maintain data privacy, federated learning (FL) trains a communal model from scattered datasets without centralized data integration. In this paper, we compare and analyze the performance of traditional deep learning (DL) and FL techniques using the chest X-Ray (CXR) image dataset for COVID-19 detection. We first implemented DL techniques VGG-16, ResNet50, and Inceptionv3, where ResNet50 is found to be best on the classification task with 98% accuracy. We then proposed FL implementations - federated averaging and federated learning using ResNet50 for training local and global models. The proposed FL converges faster and outperforms the base FL for both independent and identically distributed (IID) and non-IID datasets. While the FL handles bigger data efficiently, compared to DL, it compromised 3.56% in accuracy to preserve privacy. Our results provide a platform for the further investigation of FL in COVID-19 detection.**

Index Terms—**Chest X-ray, COVID-19, deep learning, federated learning, heterogeneity, privacy-preserving.**

I. Introduction

CORONAVIRUS disease (COVID-19) is a highly communicable disease caused by the SARS-CoV-2 virus. Although most COVID-19 patients will experience mild to moderate symptoms, COVID-19 can be deadly if left unattended even for a few days. So, early COVID-19 detection is important. The reverse transcription polymerase chain reaction (RT-PCR) test is commonly used to detect the virus's genetic material in the body due to its accuracy and reliability. The laboratory RT-PCR test requires expensive and sophisticated equipment, making it less feasible for the public. While expedient, quick, and cost-effective kits overcome the problem [1], there is a need to significantly reduce the overburdened RT-PCR laboratory testing services. Recently, medical image analysis has gained prominence in the computer-aided analysis of health conditions. AI-based analyses of medical images, such as CT and X-ray, are considered alternatives to RT-PCR for COVID-19 screening [2], [3], [4], [5]. However, massive COVID-19 screening has several limitations, for example CT scans incur higher costs and are less accessible compared to X-rays. Hence, we narrowed our research scope to X-ray image analysis. X-ray images are analyzed for COVID-19 detection and classification using deep learning (DL) implementations. A convolutional architecture, COVID-net, is one of the notable research efforts based on X-ray images [6]. In [7], [8], the authors used DL classification implementations to identify COVID-19 from other diseases; three diverse binary strategies were implemented in the former and X-ray images augmented with generative adversarial networks (GAN) to address the class data unbalancing challenge in the latter. Despite the DL research on COVID-19 achieving high detection and/or classification accuracy, it has not yet addressed data privacy at large. In the context of the current pandemic, another problem is data sufficiency. COVID-19 data is scattered across the globe and individual datasets may not be large enough to generate conclusive output. However, Big Data is a prerequisite for COVID-19 detection. In [6], [7], [9], the authors used DL to detect COVID-19, however, they faced data shortage issues, hence results are prone to bias and do not provide certain statistical confidence. Accessing and sharing COVID-19 data is prone to multiple issues, including but not limited to, sensitive and protected medical data, patient privacy policies and legal issues, etc. In such a situation, one possible solution is federated learning (FL), a technique that facilitates the building of machine learning (ML) systems without the need for centralized data. The data rests in its original site, which helps to protect privacy.

The main motivations of this study are to address the following questions:

- DL techniques are sufficiently developed to classify medical image datasets. We apply the FL technique to preserve privacy. What price do we pay to preserve privacy?
- There is a general rule of thumb that the larger the data, the better the model training.
  1) How big should the data in a medical image be?
  2) How is model efficiency affected by an incremental increase in data?
- How does parameter selection affect model efficiency?

In this paper, as depicted in Fig. 1, initially, we conduct experiments and compare traditional DL implementations and choose the best DL classification model. Additionally, we implement FL

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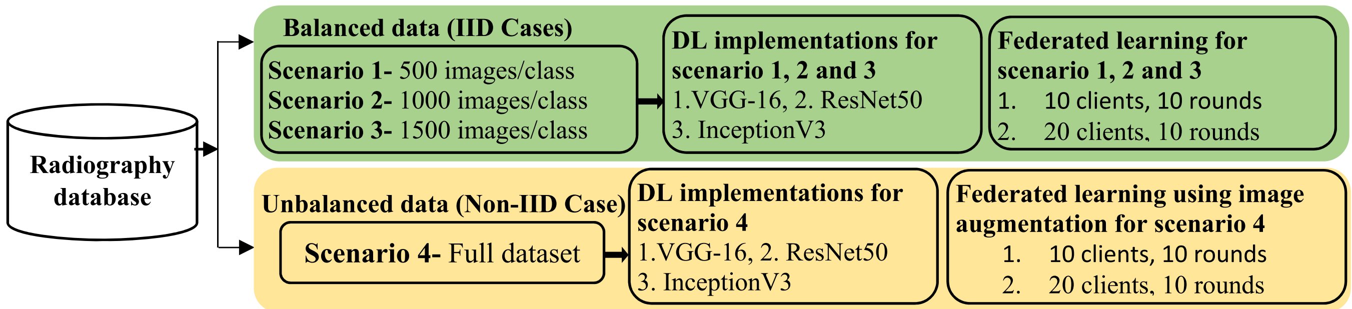


Fig. 1. Radiography COVID-19 dataset was analyzed for balanced data and unbalanced data scenarios for DL and FL implementation.

using the best DL classification model under different scenarios to analyze the CXR imaging dataset for COVID-19 detection. Furthermore, we compare the results of the proposed FL (PFL) with the base FL (BFL) implementations.

We highlight the importance of FL as a privacy-preserving technique in the health domain. This research makes the following contributions:

1) Our work differs from the previous works as we experiment with a few state-of-the-art neural network architectures as detailed in Section V-A, where ResNet50 outperforms the others, as shown in Table II in Section VI-A. We propose to use ResNet50 instead of CNN in both the FL model architecture for IID cases and the FL model architecture using image augmentation for non-IID cases. In comparison to CNN, ResNet50 overcomes the vanishing gradient problem and helps maintain a low error rate significantly deeper in the network. Moreover, ResNet50 achieves better performance due to a larger number of parameters.
2) We examine the effects of different data splits in various scenarios. The distribution of data across clients is a key challenge in FL. Hence, we apply the base federated learning approach on different data splits to examine the effects of equal and unequal data splits and their effects on COVID-19 X-ray image classification with a varying number of clients and rounds.
3) We analyse Big Data from different aspects compared to many other FL approaches. Big data analysis provides a better understanding of the data, uncovers hidden relationships, and improves findings such as classifying COVID-19 X-ray images more accurately.

The remainder of the manuscript is organized as follows. Section II discusses the existing DL and privacy-preserving research efforts in the field of medical imaging analysis, particularly for COVID-19 detection and classification. Section II gives a brief background of DL and FL techniques. Section III elaborates on the dataset used to conduct the experiments. Section IV presents an overview of the related work. In Section V, the strategies used to create the scenarios, DL and FL methods and implementations, and most importantly a comparison of the PFL implementation with BFL implementations are analyzed in Section VI. Section VII discuss the results of the experiments conducted to evaluate the efficacy of the proposed scenarios. Lastly, conclusions are drawn and future directions are discussed in Section VIII.

II. Background

In this section, we provide a brief background of DL and FL.

A. Deep Learning (DL)

Multiple pre-trained DL image classification models of varying efficiency are available. However, we limit this article to the implementation of the three most widely used pre-trained DL models which are considered state-of-the-art convolutional neural network (CNN) models for image classification; 1) VGG-16, 2) ResNet50, and 3) Inceptionv3.

The VGG-16 model achieved outstanding performance with images in extracting, distinguishing, recognizing, and classifying features [10], [11]. It is one of the pre-trained DL models most used by the research community across the globe despite new and better scoring models being available. VGG-16 stands out due to the uniform use of a 3x3 receptive field (filters) throughout the complete network with a stride of 1 pixel. In addition to the 3 convolution layers, VGG also has three non-linear activation layers, which makes the decision functions more discriminative and helps the network to converge faster. Moreover, it significantly reduces the number of weight parameters in the model and it decreases the propensity of the network to over-fit while training. Although VGG-16 is an easy and well-designed model, there are several challenges. For example, the model has more than 138M parameters and is more than 500MB in model size, which makes it more time consuming to use this model. Another challenge in using VGG is that there is no precise measure to control the gradient.

Fig. 2 presents a gradient flow that keeps flowing backward to the initial layers. This backpropagation makes the gradient smaller and significantly increases the training time.

The gradient issue was addressed in ResNet. ResNet-50 is a residual network with 50 layers. CNNs allow models to accommodate more complex functions as the number of layers increases. Therefore, the more layers, the better the performance. However, this is not a rule of thumb, as an artificial neural network no longer always delivers a substantially higher overall performance with increased hidden layers. Fewer layers help avoid overfitting in ResNet; hence we limit our analysis to

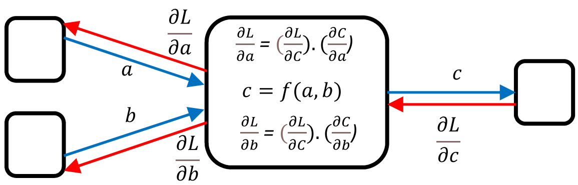


Fig. 2. Gradient flow diagram during VGG model training where $\partial L/\partial a$ and $\partial L/\partial b$ are local gradients; $\partial L/\partial z$ is a calculated gradient (back-propagating).

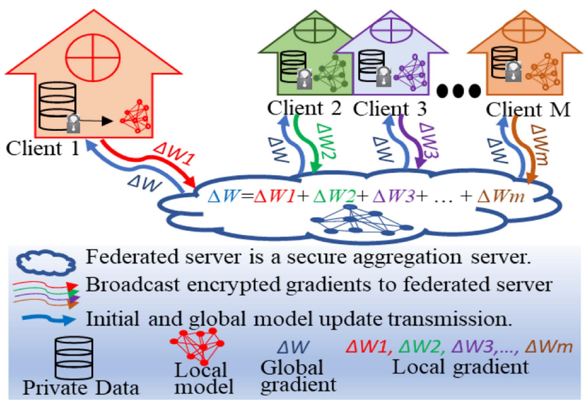


Fig. 3. Privacy-preserving FL framework. The central server initializes the model and sends the model updates/results to clients, then clients train the model updates with their local data samples and send updates back to the central server. The central server receives the updates from different clients and aggregates the results to generate the new global model. Several such rounds are executed until pre-specified stopping criteria are met.

ResNet50. ResNet allows the gradient to flow unobstructed, helping to maintain a lower error rate even in a deeper network, making it one of the most effective neural network architectures. Although ResNet is effective in lots of applications, one predominant downside is that a deeper network takes a longer training time, making it sometimes infeasible in practical applications.

The InceptionV3 model adapted numerous techniques to optimize network performance and has a deeper network than other versions of this model, such as InceptionV1 and V2, without compromising its speed. It has higher efficiency and is computationally less expensive. However, it might overfit and gives less accurate results.

B. Federated Learning (FL)

FL is an ML technique that solves privacy issues through a collaborative training implementation and does not require a single pool of centralized data. As shown in Fig. 3 and explained in [3], FL involves the following:

1) the central server initializes model training from its local data and transfers the model results to the participating clients.
2) when the clients receive the first model results from the server, each client then trains the model locally with their local data samples and returns locally trained incremental model results back to the central server.
3) the central server pools all the results and generates the global model without having any individual sample information of the clients. Here, one FL round is complete.
4) several rounds are executed in which the central server collects all the updated model results from the clients and generates the new global model. Some pre-specified criteria are set to terminate the iterative process of FL. Participating clients' data are stored and treated locally and the central server only receives updated model results.

The objective of FL is to minimize the loss function in (1). Suppose l is a client-stated global loss function achieved from a weighted combination of k local losses where $k = 1, 2, \ldots, M$ calculated from private data which is stored in the individual client's repository and is never shared between them:

$$\min_{\emptyset} l\,(X;\emptyset)\ with\, l\ (X;\ \emptyset) = \sum_{k=1}^{M} w_k l_k(X_k;\emptyset) \qquad (1)$$

where $w_k \geq 0$ represents the weights and $\sum_{k=1}^{M} w_k = 1$ [12].

III. Related Work

Several deep-learning studies have considered CXR images for COVID-19 identification [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24]. Most research work categorizes CXR images into binary or ternary classes.

Binary classifications [25], [26], [27], such as COVID and non-COVID (normal), are only discussed in a few studies, however, ternary classifications, which are COVID-19, non-COVID-19 pneumonia, and normal are used in experiments in several studies [13], [14], [16], [17], [20], [22], [24], [27], [28]. Most of the research used models which are pre-trained and readily available neural network architectures, such as VGG-16 [13], [14], [22], ResNet50 [17], [21], [22], [24], [28], Inception V3 [20], [29], while others used XGBoost [27], [30] and LightGBM [31], [32] for disease classification. ResNet50 performed relatively better and achieved the highest accuracy of ∼99%. However, different datasets with a varied number of classes and data partitions are not fit for direct comparison. To discuss comparative performance, we implemented VGG-16, ResNet50, and Inceptionv3 on the same data partitions under different scenarios. A wide range of DL experimentation was performed; however, DL does not guarantee data privacy and there is no direct solution to handle data deficiency.

FL is being used in many diverse fields, such as wireless communications, edge devices, IoT, etc. However, we focus our discussion only on medical image analysis for COVID-19 detection and classification. While FL helps to address the issues related to COVID-19 detection and classification, it faces several challenges that are briefly explained in [3], such as data heterogeneity, communication cost, and data privacy.

FL is primarily used to preserve privacy, where central servers guarantee participants remain unidentified. However, models subject to certain conditions may still remember some information. Differential privacy [33], [34] or encrypted learning implementations [35], [36] have been recommended to avoid privacy outflow in the FL framework. However, in [37], the authors indicate that full model sharing may lead to a deep privacy breach [37], [38].

Communication efficiency during FL rounds is addressed in various studies [39], [40], however, it is still a bottleneck.

We limit our work for communication efficiency and privacy which, by default, an FL framework holds. This work studies different data split effects on model performance, where we experiment with FL using ResNet50 with different parameters. We also study the efficiency vs privacy tradeoff. Experiments were conducted with FL implemented on CXR image data in IID and non-IID settings in [34], [41]. To handle the heterogeneity of data, CXR image scaling and augmentation techniques are adopted for data pre-processing in [42], [43] for binomial and multiclass data respectively before applying FL for COVID-19 detection. The former implementation may produce ambiguous results which fall into one of the two categories, namely True or False, however, the latter technique provides a better understanding of the data and results in better screening. We experiment with multiclass data. Image augmentation is also used to reduce communication costs in [41] to cater for non-IID cases to deal efficiently with real-world data. Likewise, in [34], the author implemented generative adversarial networks (GAN) in conjunction with FL to diagnose COVID-19 for non-IID data and successfully maintained privacy.

We use the BFL technique presented in [40], [41] where the authors experiment with only one scenario and distribute data to 20 clients to study IID and non-IID model outputs. However, we conducted a series of experiments with FL implementations using ResNet50.

IV. COVID-19 Dataset

The COVID-19 Radiography Database currently comprises CXR images with images of positive COVID-19 cases along with Normal, Lung Opacity, and Viral Pneumonia images. The COVID-19 dataset consists of images which are 256x256 in size and are in portable graphics format (PNG). This database and ongoing research are executed to examine the efficacy of AI in the secure detection of COVID-19 from CXR images. In total, the COVID-19 dataset from the COVID-19 radiography database (https://www.kaggle.com/tawsifurrahman/covid19-radiography-database) consists of 21165 samples divided into four classes: Covid-19 (C) 3616, Normal (N) 10192, Lung-Opacity (L) 6012 and Viral Pneumonia (VP) 1345 images. The currently available COVID-19 dataset contains enough images and does not require data cleansing. The dataset does not contain any duplicate or null samples.

This work analyzes data under different scenarios consisting of 500, 1000, and 1500 images per class for IID settings, so we refrained from using the Viral Pneumonia dataset, which only has 6.4% of the total data containing 1345 samples. Only data for ternary classes i.e., COVID-19, Normal, and Lung-Opacity are analyzed.

V. Methods

We explain the DL methods followed by FL.

A. Centralized Learning Approaches

In this section, we outline the details of the DL methods. We employed traditional DL techniques on IID and non-IID data.

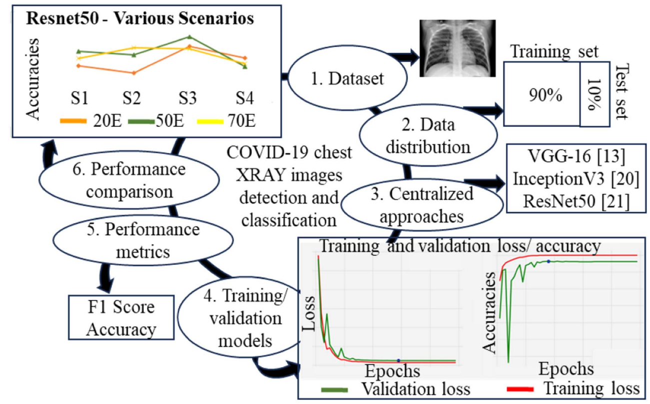


Fig. 4. Architectural design for DL analysis: First, the dataset (for each scenario) is distributed into two parts, 90% and 10% ratio for training and testing purposes respectively. After that, centralized approaches using pretrained DL models were used to observe their training and validation accuracy and loss plots to understand the significance of the models. Then performance metrics are used for the classification of the images (COVID-19 XRAY images detection and classification), and finally, centralized approaches performances are compared for various number of epochs.

1) Centralized Architectural Design: In this work, we implemented 3 centralized pre-trained and readily available neural network architectures VGG-16, ResNet50, and Inceptionv3. Detailed implementations of DL approaches are out of the scope of our paper. We designed to execute our DL experiments for COVID-19 XRAY images detection and classification are presented in Fig. 4.

We used Keras and TensorFlow for the experimentation. Keras is a powerful, user-friendly, open-source software library that provides a Python interface for artificial neural networks, and TensorFlow is an open-source library to perform machine learning tasks using Keras as the backend. We trained DL models on GPU NVIDIA-SMI 470.57.02, driver version: 470.57.02 CUDA version: 11.4.

2) DL Method and Implementation: We used a 224×224 input image size which is the default image size of VGG-16 and ResNet50, however, the default input image size of Inception-v3 is 299×299. We train the inceptionV3 model on PyTorch which uses an adaptive average pooling layer right before the fully connected layer. As a result, the inputs to the fully connected layers are always the same size, so no exceptions are noted. As shown in Fig. 5, DL is implemented as follows:

1) Prepare input data: Start preparing input data for IID cases. Prepare input data for scenario 1 by preparing a folder with the first 500 images from each class. Then, add 500 more images per class to prepare a data folder for scenario 2 where each class consists of 1000 images. Follow the same practice and add 500 more images per class to prepare a data folder for scenario 3 where each class consists of 1500 images.
2) Split training and testing images.
3) Apply pre-trained DL models, i.e., VGG-16, ResNet50, and Inceptionv3.
4) Prepare outputs for different parameters for each scenario.
5) Compare model accuracies under different scenarios.

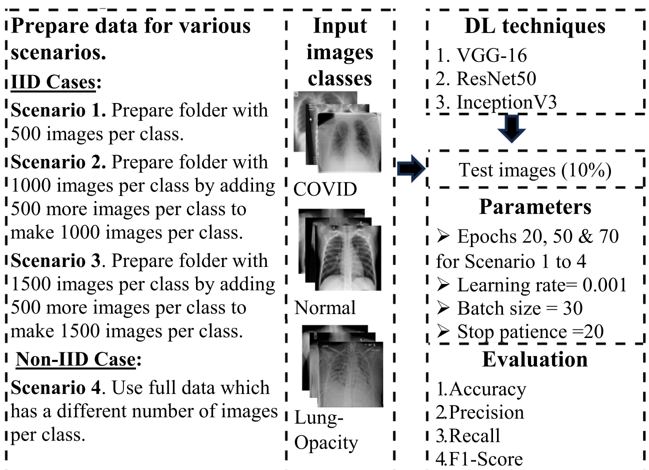


Fig. 5. Workflow of DL analysis implementations under various scenarios. Input images consist of 3 classes, COVID, Lung-Opacity, and Normal. 10% of the data was used for testing purposes. The DL techniques, parameters and evaluation criteria are given.

Our selected DL models are pre-trained on ImageNet, a project which aims to provide a large image database for research purposes and is used as a starting point for model training. The max pooling layer is added to decrease the dimensions of the feature maps and helps in extracting low-level features like edges, points, etc. The softmax activation function is used in the model output layer to predict a multinomial probability distribution to help the network learn complex patterns in the data. We used an intuitive method called early stopping to choose the number of DL model iterations to determine when to stop the model training process which depends on the validation error. The early stopping technique stops the model training if the validation error has not improved over the previously specified number of epochs, say 20 in our case (stop patience = 20). Adamax, an extension to the Adaptive Movement estimation (Adam) version of gradient descent, is used to optimize the model weights. Data is unbalanced when more than half of the data belongs to the normal class. However, we analyze data in 3 IID and 1 non-IID cases, and we considered accuracy a suitable measure to compare model performances for IID cases. For non-IID cases, accuracy alone is not a good measure to compare DL classification model performances. After data balancing through image augmentation, average accuracy is one suitable measure, however, the weighted average of F1 is a better metric for imbalanced data to compare model performances in the case of ternary data classification. Early stopping criteria and data augmentation techniques also helped to avoid model overfitting.

To check the statistical significance of the DL approaches, a non-parametric test to compare two samples with unequal variances, two-tailed Welche's t-test was applied. The lowest false negative rate in cases like COVID-19 is desirable because we do not want COVID positive cases classified as negative to avoid the risk of COVID spread. The significance of the model performance is not measured only by accuracy but average accuracy metrics such as precision, recall, and F1-score are also computed to obtain deeper information about the classifier using a succinct method, the confusion matrix. The confusion matrix enables a detailed model analysis so a fair conclusion can be drawn about its performance.

The average accuracy metric and weighted average of F1 are computed as follows:

Let us consider three classes Normal (N), Lung- Opacity (L), and COVID (C), and

$$\mathrm{TP} = \mathrm{True\,Positive},\ \mathrm{FP} = \mathrm{False\,Positive}$$
$$\mathrm{TN} = \mathrm{True\,Negative},\ \mathrm{FN} = \mathrm{False\,Negative}$$

Average accuracy:

$$\mathrm{A}\,(\mathrm{N}) = \frac{TP\ of\ N + TN\ of\ N}{TP\ of\ N + TN\ of\ N + FP\ of\ N + FN\ of\ N}$$
$$\mathrm{A}\,(\mathrm{N}) = \frac{TP\ of\ L + TN\ of\ L}{TP\ of\ L + TN\ of\ L + FP\ of\ L + FN\ of\ L},$$
$$\mathrm{A}\,(\mathrm{C}) = \frac{TP\ of\ C + TN\ of\ C}{TP\ of\ C + TN\ of\ C + FP\ of\ C + FN\ of\ C}$$
$$\mathrm{Average\ accuracy} = \frac{A(N) + A(L) + A(C)}{No.\ of\ classes}$$

Average Precision:

$$\mathrm{P}\,(\mathrm{N}) = \frac{TP\ of\ N}{TP\ of\ N + FP\ of\ N},\ \mathrm{P}\,(\mathrm{L}) = \frac{TP\ of\ L}{TP\ of\ L + FP\ of\ L}$$
$$\mathrm{P}\,(\mathrm{C}) = \frac{TP\ of\ C}{TP\ of\ C + FP\ of\ C},\ \mathrm{Average\ precision} = \frac{P(N) + P(L) + P(C)}{No.\ of\ classes}$$
$$\mathrm{Average\ Recall} : \mathrm{R}\,(\mathrm{N}) = \frac{TP\ of\ N}{TP\ of\ N + FN\ of\ N},$$
$$\mathrm{R}\,(\mathrm{L}) = \frac{TP\ of\ L}{TP\ of\ L + FN\ of\ L}$$
$$\mathrm{R}\,(\mathrm{C}) = \frac{TP\ of\ C}{TP\ of\ C + FN\ of\ C},\ \mathrm{Averagerecall} = \frac{R(N) + R(L) + R(C)}{No.\ of\ classes}$$
$$\mathrm{F1} = 2 * \frac{Average\ Precision * Average\ Recall}{(Average\ Precision + Average\ Recall)}$$

Here, F1 is the harmonic mean of average precision and average recall. DL model testing average accuracies and macro averages of precision, recall, and F1 are reported in Table II.

B. *Federated Learning Approaches*

In this section, we outline the details of the FL methods. we explain the BFL model architecture and implementations for both IID and non-IID cases followed by the PFL model architecture and implementation.

1) FL Architectural Design: In this research work, we explain the FL architectural design to execute FL experiments presented in Fig. 6.

FL can be implemented through various frameworks, such as Tensorflow, PySyft, PyTorch [48], ImageNet, and Scratch [42], etc. which require, including but not limited to, configuration in the local machines and heavy GPUs. Google Colaboratory or Google Colab (GC) for short is an open-source framework on which to run Python codes in a local browser with zero configuration. GC is efficient once connected with GPU. We implemented FL techniques on GC PRO GPU NVIDIA-SMI 460.32.03, driver version: 460.32.03, and CUDA version: 11.2.

2) FL Method and Implementation: FL analysis workflow explained in Fig. 7 is followed by both BFL and PFL techniques for IID and non-IID cases.

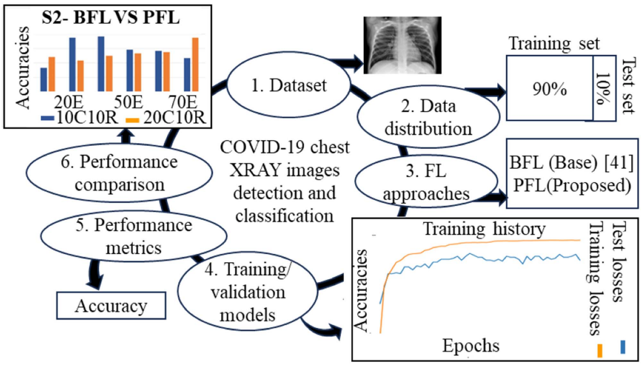


Fig. 6. Architectural design for FL analysis: First, the dataset (for each scenario) is distributed into two parts, 90% and 10% ratio for training and testing purposes respectively. After that, FL approaches were used to observe their training and validation accuracy and loss plots to understand the significance of the models. Then performance metrics are used for the classification of the images (COVID-19 XRAY images detection and classification), and finally, FL approaches performances are compared for various number of clients, rounds and epochs.

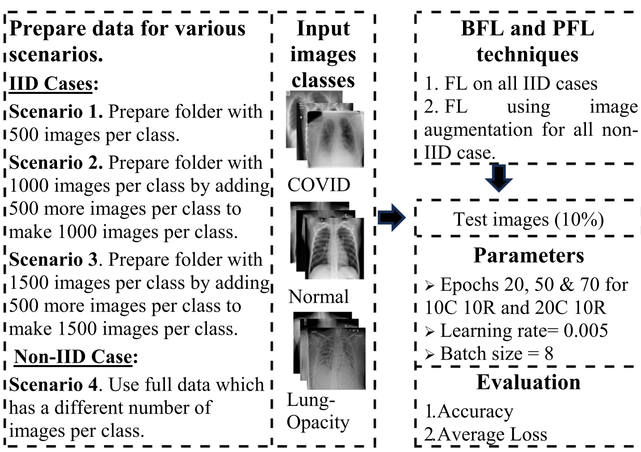


Fig. 7. Workflow of FL analysis for base and proposed implementations under various scenarios. Input images consist of 3 classes, COVID, Lung-Opacity, and Normal. 10% of the data was used for model testing. The DL and FL methods, their corresponding parameters and evaluation criteria are given.

We will discuss BFL methodology and implementation details for IID and non-IID cases followed by PFL.

a) BFL Method and Implementation: a. BFL For IID Cases

FL requires a DL model to execute an FL process. The DL model is used at the server location to train the initial and global models, and each client trains the DL model with their local data after receiving the updates from the central server. The BFL implementation employs the CNN model proposed in [40] shown in Fig. 8. CNN stands out from the class of DL models due to its high accuracy when used for image classification and recognition. A traditional CNN model consists of 1) convolutional (Conv) layers for feature extraction, 2) pooling layers responsible for the reduction in the number of features and parameters, and 3) fully connected layers to execute mapping amongst the extracted features and the ultimate results. A referenced CNN architecture, as shown in Fig. 8, consists of 5 Conv layers with kernel sizes 7×7, 5×5, and 3×3 for the other three, subsequently 2 fully connected layers and for each Conv layer 2×2 maxpooling and dropout layers are added to overcome the overfitting issues. Moreover, at the end of the CNN architecture [40], the rectified linear unit (ReLU) activation function is used for Conv and the fully connected layers. For FL implementation, we revisit Fig. 3 and explain the BFL method as follows:

1) Users receive initial CNN model updates ω_t from the server and train a new CNN model using updates and local samples and return the locally trained CNN model incremental results ω_{r+1}^{m} back to the central server.
2) The central server pools all the model updates received from different clients by the FedAvg method stated in (2) and generates the enhanced global CNN model without having any individual sample information of the users to complete one FL learning round.

$$\omega_{r+1}^{m} \leftarrow user\ update\ (\omega_r,\ m)\ and\ \sum_{m=1}^{M} \frac{n_m}{n}\omega_{r+1}^{m} \quad (2)$$

3) Repeat steps 1 to 3 until a pre-set threshold is met.

Data is distributed randomly to each client. The CNN used for the FedAvg algorithm contains increased layers adding dimensionality and decreasing noise in an image which helps improve neural network performance. FL efficiency is evaluated based on testing the accuracies reported in Table III.

b. BFL For non-IID Cases.

The data augmentation technique uses image cropping, padding, flipping, etc. strategies to upsurge the quantity of image data and helps address non-IID data with a varied number of images per class. This technique makes the model more vigorous and avoids overfitting.

Here, the base architecture implements FL using the image augmentation technique anticipated in [41] for non-IID data scenarios, where the authors propose a novel implementation to avoid a performance drop due to non-IID image data cases.

As mentioned in Section IV, the unbalanced data consists of 3616 COVID-19, 10192 Normal, and 6012 Lung-Opacity images. In [41], the authors implemented image augmentation to balance the local data distribution. Base model architecture, the CNN model explained in Fig. 8, is used for FL implementation. Dirichlet distribution is used for central data distribution to different clients in a non-IID manner.

The probability density function (pdf) of Dirichlet distribution is shown in (3).

$$f\ (v_1,\ v,\ \ldots v_m;\ p_1,\ p_2, \ldots,\ p_m) = \frac{1}{B\,(p)}\ \pi_{j=1}^{m}\ v_i^{(p_i-1)} \quad (3)$$

where $B\ (p) = \frac{\pi_{j\,=\,1}^{m}\ r(p_i)}{r(\sum_{i\,=\,1}^{m} pi)}$ and $\sum_{i\,=\,1}^{m} v_i$

Suppose there are M clients, where $v_1,\ v,\ \ldots v_M > 0$ and $p_1,\ p_2, \ldots,\ p_M$ are sampled variables and parameters of the distribution respectively. It can be seen from (3) that the imbalance level of sampled values v increases with an equivalent increase in p parameters.

The work in [41] experiments with distributed data on various clients under different scenarios. Let us consider one of the scenarios and artificially distribute data to 10 individual clients

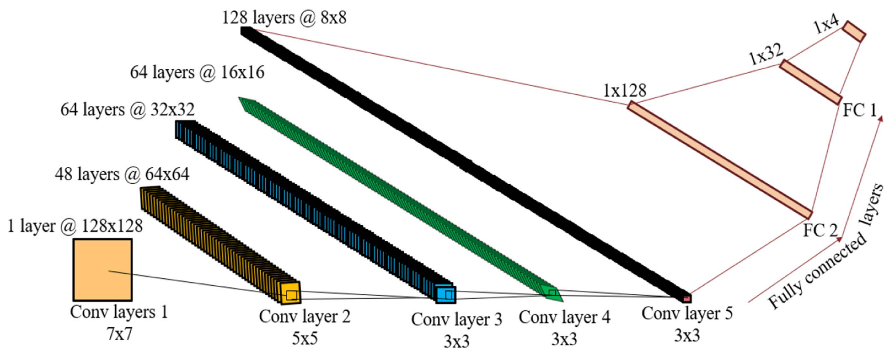


Fig. 8. Base CNN architecture used to classify multi-chest diseases.

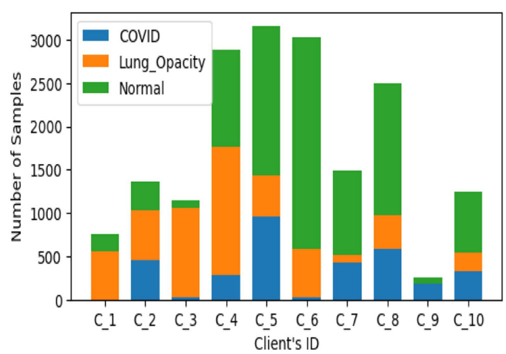


Fig. 9. Local data distribution of the clients in a non-IID scenario. Each client contains a different number of CXR images from each class.

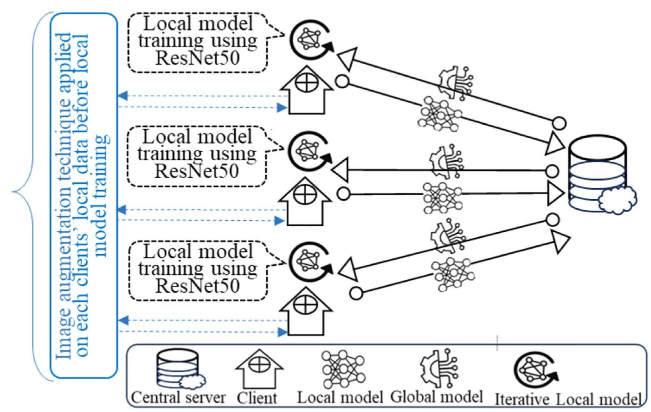


Fig. 10. CXR images classification PFL model training for IID (black) and non-IID (black and blue) cases. For IID model training (black), central server sends the model updates to the clients and train a new local model and new local model updates will be sent back to the central server to complete one FL round. However, for non-IID model training one extra step, image augmentation (blue) is added for each client to balance data distribution and avoid overfitting.

in a non-IID fashion. The non-IID partitioning of the central dataset into local data clients is presented in Fig. 9.

To balance the distribution of the chosen clients, the authors in [41] proposed a technique where the image augmentation phase is added to the vanilla FedAvg algorithm. Synthetic images are generated using various transformations, such as horizontal and vertical flips, crop, invert, solarize, rotation, jitter, perspective, sharpness, Gaussian noise, histogram equalization, contrast, Gaussian blur, and affine. After the augmentation phase, inter and intra-client

3) PFL Method and Implementation: a) PFL For IID Case

The prime goal of the classification model is to classify images precisely. We experimented with a few state-of-the-art neural network architectures as detailed in Section V (1), where ResNet50 performed best. We propose to use ResNet50 instead of CNN in both the BFL model architecture for IID cases and the BFL model architecture using image augmentation for non-IID cases.

In the FL iterative process, we propose to use ResNet50 for local model training. As shown in Fig. 10, an initial or global model will be sent by the central server to a client. The client will use its local data and initial or global model updates and train a new model using ResNet50. Then a locally trained new model will be sent back to the central server. All locally trained model updates from various clients are aggregated at the central server. There are several aggregation implementations, however, FedAvg permits local users to execute more than one batch update on local data and exchange the updated weights instead of gradients, which is considered the most suitable for the analysis.

For the FL implementation, we consider Fig. 10 and also revisit Fig. 3 to explain the PFL method as follows:

1) The central server initializes the ResNet50 model and sends the model updates ω_t to M users.
2) Users receive initial ResNet50 model updates ω_r from the server and train the new ResNet50 model using updates and local samples and return locally trained ResNet50 model incremental results ω_{r+1}^{m} back to the central server.
3) The central server pools all the model updates received from different clients using the FedAvg method stated in (2) and generates the enhanced global ResNet50 model without any individual sample information of the users to complete one FL learning round.

$$\omega_{r+1}^{m} \leftarrow user\ update\ (\omega_r,\ m)\ and\ \sum_{m=1}^{M} \frac{n_m}{n} \omega_{r+1}^{m} \quad (4)$$

4) Repeat steps 1 to 3 until a pre-set threshold is met.

Data is distributed randomly to each client. PFL for IID case efficiency is evaluated based on the testing accuracies reported in Table IV.

TABLE I
MAIN PARAMETERS AND METRICS

Parameter	Value
Clients	10, 20
Rounds	10
Epochs	20,50, 70
Local epochs	10
Learning rate	5x10^{-3}
Batch size	8
Optimizer	Adadelta
Performance metric	Cross entropy loss

When FL coding, we used a powerful function cross-entropy to measure the probability of a model. Cross entropy is a loss function often used in the classification problems which describes how likely a model is and the error funtion of each data point, making it a very useful function. It is also used to compare a predicted outcome compared to the true outcome.

b) PFL For Non-IID Cases As explained in BFL non-IID case (see Section V-B.1(b)), data augmentation technique is implemented in PFL for non-IID case to balance data distribution and helps avoid model overfitting. Non-IID data is randomly distributed to each client. On each client's unbalanced data, we implement image augmentation to balance the local data distribution. All the technical details are mentioned in Section V-B.1(b).

In the FL iterative process, we propose to use ResNet50 for local model training. As shown in Fig. 10, an initial or global model will be sent by the central server to a client. The client will use its new balanced local data, and initial or global model updates and train a new model using ResNet50. Then a locally trained new model will be sent back to the central server.

The proposed model architecture, the ResNet50 model is used for FL implementation. Dirichlet distribution is used for central data distribution to different clients in a non-IID manner. For the FL implementation, we consider Fig. 10 and also revisit Fig. 3 to explain the PFL method for non-IID cases as already explained in PFL for IID cases (see Section V-V.2(a)) except for each client we implement image augmentation technique to balance the local data distribution. Then initial or global model updates will be jointly trained with augmented data to get new local model. Then local model updates will be sent back to the central server to complete one FL round. PFL for non-IID case efficiency is evaluated based on the testing accuracies reported in Table IV.

4). Model Parameters and Metrics: The BFL and PFL experiments are conducted using the following parametric values and metrics for all IID and non-IID cases presented in Table I.

VI. EXPERIMENT RESULTS

This section presents the experiment results. Initially, we discuss the results of the DL implementations, followed by the results and a comparison of the BFL and PFL implementations.

A. DL Results

DL implementations were trained for a various number of epochs; 5, 20, 50, and 70. The DL models produced very promising results, as shown in Table II. However, the unstable results of VGG-16 and the results of all the DL models for 5 epochs are not considered for this analysis therefore, they are not reported in Table II. From the overall DL analysis, we draw the following conclusions:

1) Considering the overall performance results, it has been seen that the pre-trained ResNet50 model outperforms the others and achieves the highest accuracy of 98% for 50 epochs under scenario 3.
2) Interestingly, ResNet50 for 20 epochs and Inception V3 for 50 epochs produced the same results in 2 of 3 IID cases. Hence, we can conclude that ResNet50 converges faster and is a relatively efficient DL implementation.
3) We use the performance metric F1 score for non-IID cases. The higher the F1 score, the better the model performance. In non-IID cases, ResNet50 and InceptionV3 performed better under 50 and 70 epochs.
4) Three points are very clear from Fig. 11 for all three models. First, the results under scenario 1 are quite unstable, whereas the results under Scenario 3 are relatively better. Second, an increase in the number of epochs may not always increase model performance. As in most of the results for the epochs, the results for 70 epochs are the same as 50 epochs or less. It can be clearly seen from Fig. 11 that the model performance for 50epochs is the most stable. Third, we observe that non-IID data behavior, which is scenario 4 in each figure, shows very stable model performance for all three models.

The last column and last row of Table II shows the p-values of the two-tailed Welche's t-test for scenarios and epochs respectively comparing statistical significance of ResNet50 and InceptionV3. In all comparisons, there's not enough evidence to reject null hypothesis on the tests with 95% confidence level (p-values$>$0.05), giving statistically non-significant results. In this situation, statistical tests are unable to conclude that one DL method is better than the other in a meaningful way. Still, there are several points to consider justifying comparison such as performance metrics, cross validation, computational efficiency, etc. However, accuracies show that all the chosen DL approaches are performing very well. In such closely competitive approaches consistent results (all $>$90%) of Table II across different data splits cross validate and add credibility to our findings. Class-wise metrics assist in comprehending how well each model achieves in individual classes. ResNet50 is not only theoretically favourable (see Section II(a)) but also gives least misclassification in case of COVID-19 under all scenarios.

Fig. 12(a)–(c) present the errors by class plot, the confusion matrix and the classification report of the ResNet50 model performance under scenario 2. Only 3, 4 and 3 images are classified as incorrect from the Normal, Lung-Opacity and COVID images out of 150 images (test set) for each class. We can visualize the actual and predicted values from the confusion matrix and accuracy metrics precision, recall and f1-score. F1-score uses both precision and recall, which tells us that 98% of the positive predictions made by the model are correct. We computed the classification reports for all the models under each scenario, and the F1 scores are reported in Table II.

Hence, we conclude from the DL experimentation that ResNet50 for 50 epochs is the most suitable model for

TABLE II
OVERALL PERFORMANCE RESULTS OF THE EVALUATED DL TECHNIQUES UNDER VARIOUS SCENARIOS

	DL implementations	Average accuracies (F1 Score)			p-value
		20 Epochs	50 Epochs	70 Epochs	
Scenario 1 – 500 images/class	ResNet50	**95.33%** (0.95)	96.67% (0.97)	96.00% (0.96)	0.1332
	Inception V3	93.33% (0.93)	**95.33%** (0.95)	94.67% (0.95)	
Scenario 2 – 1000 images/class	ResNet50	94.67% (0.95)	96.33% (0.96)	97.00% (0.97)	0.2254
	Inception V3	96.67% (0.97)	97.00% (0.97)	96.67% (0.97)	
Scenario 3 – 1500 images/class	ResNet50	**97.11%** (0.97)	**98.00%** (0.98)	96.89% (0.97)	0.1336
	Inception V3	95.11% (0.95)	**97.11%** (0.97)	96.44% (0.96)	
Scenario 4 – Non-IID case (All images included)	ResNet50	96.06% (0.96)	95.26% (0.96)	95.51% (0.96)	0.4226
	Inception V3	95.21% (0.95)	95.46% (0.96)	95.91% (0.96)	
	p-value	0.4415	0.6631	0.4897	

The highest accuracy are highlighted in bold values.

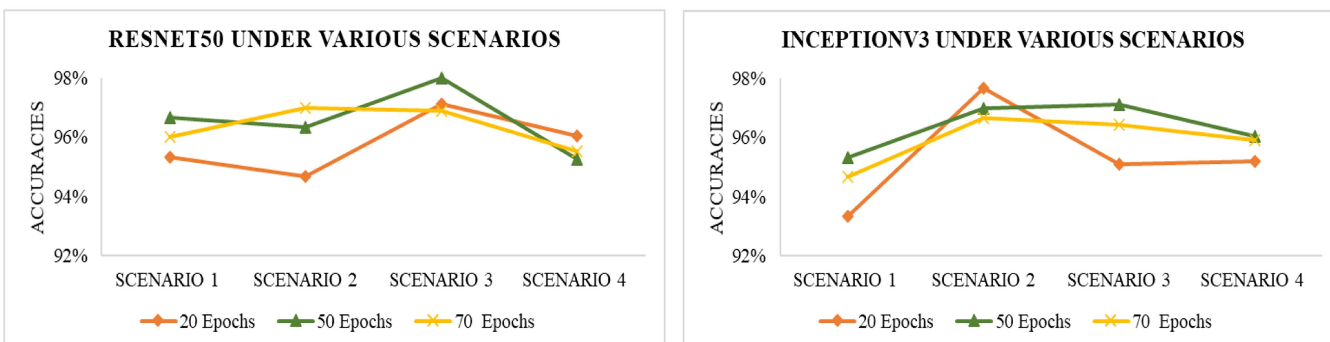
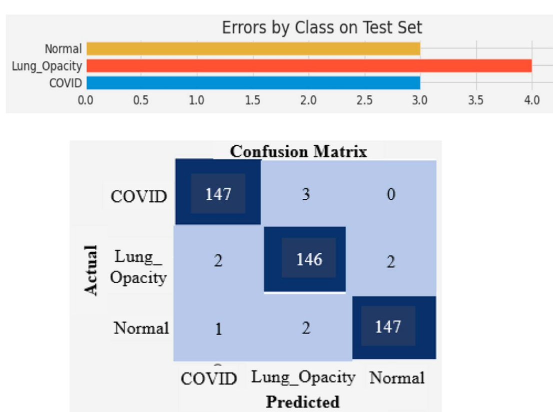


Fig. 11. DL model (RESNET50 and Inception V3) performances under various scenarios. The various scenarios are Scenario 1 – 500 images per class, Scenario 2 – 1000 images per class, Scenario 3 – 1500 images per class, and Scenario 4 – Non-IID case (All images included).

(a)

(b)



(c) Classification Report:

	precision	recall	f1-score	Support
COVID	0.98	0.98	0.98	150
Lung_Opacity	0.97	0.97	0.97	150
Normal	0.99	0.98	0.98	150
accuracy			0.98	450
macro avg	0.98	0.98	0.98	450
weighted avg	0.98	0.98	0.98	450

Fig. 12. ResNet50 outcomes for 50 epochs under scenario 2 are displayed in (a) error by class on test dataset, (b) confusion matrix and (c) classification report.

COVID-19 Radiography Database CXR image classification. Scenario 3, which involves 1500 images per class and a total of 4500 images resulted in the best overall performance for DL.

B. FL Results

Experiments were conducted using FL implementations for a various number of epochs: 5, 20, 50, and 70 and also for a varied number of clients and rounds. This section discusses the results of BFL and the PFL provided in Table III and Table IV, respectively. However, the unstable results of both FL implementations for 5 epochs are not considered for analysis and are therefore not reported in Tables III and IV.

1) BFL Model Results: The BFL model results are presented in Table III. From the overall BFL analysis, we draw the following conclusions:

1) The BFL implementation outperformed for 10C10R with 50 epochs under scenario 2.
2) The BFL IID case results presented in Table III portray a similar output pattern as DL, e.g., a decrease in test accuracy for 70 epochs compared to 50 in scenarios 1 and 2.
3) Unlike DL, the overall BFL results for 50 epochs are slightly better for scenario 2 compared to 3.
4) When comparing a varied number of clients and rounds, the overall performance of scenario 2 is relatively better.
5) A steady increase in accuracy for the FedAug implementation for non-IID can be observed in Table III.

We conclude from the BFL experiments that 10C10R for 50 epochs is the most suitable for the COVID-19 Radiography Database CXR image classification. The BFL implementation

TABLE III
OVERALL PERFORMANCE RESULTS OF THE EVALUATED BFL IMPLEMENTATION UNDER VARIOUS SCENARIOS

	FL implementations	**20 Epochs**	**50 Epochs**	**70 Epochs**
Scenario 1 – 500 images/class	10 clients, 10 rounds (10C10R)	90.00%	86.67%	85.33%
	20 clients, 10 rounds (20C10R)	82.00%	88.67%	85.33%
Scenario 2 – 1000 images/class	10 clients, 10 rounds (10C10R)	86.67%	**95.33%**	91.33%
	20 clients, 10 rounds (20C10R)	89.67%	90.00%	<u>91.00%</u>
Scenario 3 – 1500 images/class	10 clients, 10 rounds (10C10R)	88.89%	91.78%	93.78%
	20 clients, 10 rounds (20C10R)	89.33%	90.67%	88.89%
Scenario 4 – Non-IID case (All images included)	10 clients, 10 rounds (10C10R)	86.02%	87.29%	88.65%
	20 clients, 10 rounds (20C10R)	85.57%	86.18%	89.20%

The highest accuracy are highlighted in bold values.

TABLE IV
OVERALL PERFORMANCE RESULTS OF THE EVALUATED PFL IMPLEMENTATION UNDER VARIOUS SCENARIOS

	FL implementations	**20 Epochs**	**50 Epochs**	**70 Epochs**
Scenario 1 – 500 images/class	10 clients, 10 rounds (10C10R)	85.33%	84.67%	85.33%
	20 clients, 10 rounds (20C10R)	81.33%	88.00%	91.33%
Scenario 2 – 1000 images/class	10 clients, 10 rounds (10C10R)	**95.00%**	91.67%	89.33%
	20 clients, 10 rounds (20C10R)	88.67%	90.67%	**95.00%**
Scenario 3 – 1500 images/class	10 clients, 10 rounds (10C10R)	91.33%	**94.44%**	92.00%
	20 clients, 10 rounds (20C10R)	88.67%	88.89%	91.56%
Scenario 4 – Non-IID case (All images included)	10 clients, 10 rounds (10C10R)	88.55%	**91.02%**	90.67%
	20 clients, 10 rounds (20C10R)	88.09%	89.81%	90.77%

The highest accuracy are highlighted in bold values.

achieved the best results for Scenario 2, 1000 images per class, and a total of 3000 images.

2) PFL Model Results: The PFL model results are presented in Table IV. From the overall PFL analysis, we draw the following conclusions:

1) The PFL implementation outperformed for 10C10R for 20 epochs and 20C10R for 70 epochs under scenario 2.
2) In the case of bigger data, such as scenario 3, the PFL implementation also produced promising results, particularly under 50 epochs.
3) From both scenario 2 and scenario 3, the 10C10R data split performed relatively better. However, scenario 2 with 1000 images per class produced the best results for 20 epochs, where scenario 3 with 1500 images showed similar accuracy for 50 epochs.
4) In the PFL IID and non-IID cases, an overall steady increase in accuracy with an increase in the number of epochs can be observed from Table IV.

We conclude from the PFL experiments that the 10C10R data split is the most suitable for COVID-19 Radiography Database CXR image classification. Overall, the PFL implementation handles bigger data with similar accuracy with an increased number of epochs.

C. Comparison- BFL vs. PFL Model Results

We draw the following conclusions from a comparison of the BFL and PFL model results:

1) The PFL implementation achieved 95% accuracy for 20 epochs, as presented in Table IV, exhibiting faster convergence than the base implementation which achieved 95.33% accuracy in 50 epochs, as shown in Table III.

From the results of scenario 2 in Fig. 13, Tables III, and IV, we can see the promising results of the PFL compared to the BFL under 20 and 70 epochs.

From the results of scenario 3, Tables III, and IV, we can conclude that the PFL implementation handles bigger data efficiently. Fig. 13 shows that the PFL achieves higher accuracy than the BFL.

2) Overall, the PFL performed better than the BFL for a data split of 10 clients and 10 rounds (10C10R) for almost each epoch under all scenarios, as shown in Fig. 13.

For the non-IID case, scenario 4 in Tables III and IV show a steady increase in accuracy. The PFL implementation achieves improved results for an increased number of epochs, which leads us to train the model for even higher epochs which may produce better results.

3) For the non-IID case, scenario 4, the PFL achieved better performance, as shown in Fig. 13.

In conclusion, the PFL implementation can handle bigger data efficiently and a 10C10R data split with 50 epochs under scenario 3 achieved the best performance and is the most suitable for the COVID-19 Radiography Database CXR image classification.

A plot of DL vs BFL vs PFL is displayed in Fig. 14. We compared the outputs of the DL vs BFL and PFL for 10C10R under scenario 3. The DL ResNet50 with the IID setting performed the classification task on a 10% test set with 98% accuracy after 50 epochs under scenario 3, whereas BFL and PFL achieved a 91.78% and 94.44% accuracy respectively under the same scenario. The DL technique ResNet50 achieved the highest accuracy but did not preserve privacy. However, the FL techniques preserve privacy by default and the proposed technique performed 2.66% better than BFL. In this situation, to preserve

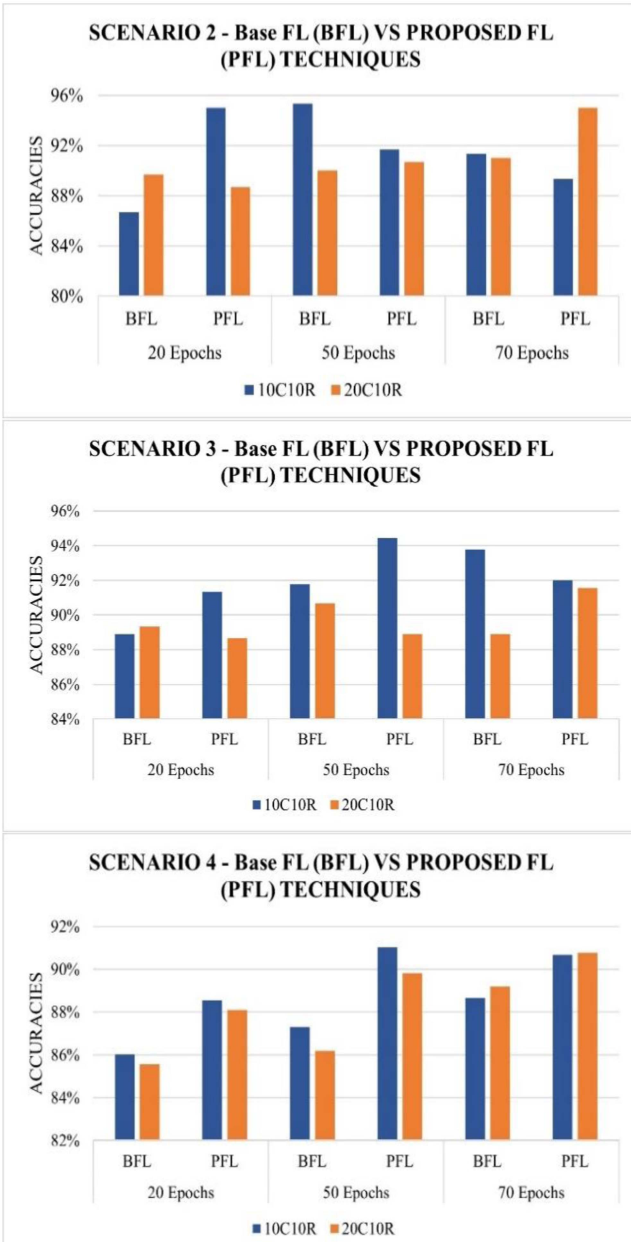


Fig. 13. Performance of the BFL and PFL for 20, 50 and 70 epochs for various clients under IID scenarios 2 and 3; and non-IID scenario 4. The various clients and rounds of federated training are 1) 10 Clients,10 Rounds (10C10R), and 2) 20 Clients,10 Rounds (20C10R); and Scenario 2 – 1000 images/class, Scenario 3 – 1500 images/class, and Scenario 4 – Non-IID case (All images included).

privacy, our proposed implementation only compromises 3.56% accuracy in comparison to the DL implementation. Hence, these results show that after the careful selection of the parameters, our PFL technique is a good alternative to traditional DL and performs better than BFL.

VII. Result Discussion

When model training, we experimented to identify and fine-tune the most effectual parameters to attain the best accuracy with minimal loss. We fine-tuned the models using different

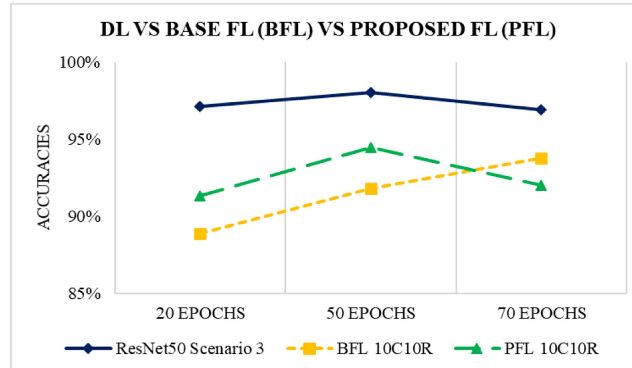


Fig. 14. Comparison plot between DL scenario 3 outputs vs BFL and PFL for 10C10R under scenario 3 results.

values of clients, rounds, epochs, learning rates, data slices and batch sizes. We recorded the effects of the abovementioned parameters and found the best values as detailed in Table I. In addition, an Adadelta optimizer was used to fine-tune the models. We also learned that FedAvg achieves better testing accuracy and loss, changes in the learning rate have a trivial effect on model testing accuracy but an increase in the data size and the number of epochs has a significant effect, however a very large number of epochs (i.e., >70) did not have any noticeable effect on testing accuracy.

We also compared our experiment results with other FL approaches analysing COVID-19 medical images in Table V. In the work in [42], [49], [50], the authors ran extensive rounds which may consume comparatively more execution time. The work in [40], [41], [42], [48], [49], [50] report comparatively lower performance, however the work in [51] performs better but only classifies binary data. The work in [52] focuses on reducing the training time but running less epochs may result in lower accuracy which is not reported clearly in the paper. Studies such as [40], [41], [42], [51], [52], [53] implement FL on medical images to detect or classify COVID-19 where data is only distributed either in 2 or 3 clients. In [48] and [49], the authors collaborated with 5 and 10 clients respectively, however the results were lower than our proposed approach. In our approach, we collaborate with more clients and not only perform efficiently in IID cases but also in non-IID cases.

VIII. Conclusion and Future Directions

FL is a method to collaboratively learn various users' data without the need to centralize data for analysis and is gaining popularity, particularly in fields where data privacy is of critical importance and data breaches may have legal implications in the future, such as the health sector, financial sector, etc.

We conducted experiments on one of the largest and most prominent datasets available, the Radiography CXR images dataset from COVID-19 patients [45], [47]. The data details are discussed in Section IV. The test dataset consists of 10% of the total dataset for each scenario. We experimented with DL, BFL and PFL implementations under various scenarios. We draw the following conclusions from the results of the experiments:

TABLE V
RESULTS OBTAINED COMPARED TO OTHER FL APPROACHES ANALYSING COVID-19 MEDICAL IMAGES. (C: CLIENTS, R: ROUNDS)

Ref.	DL model used for FL	Tasks	Data Source	C	R	Performance
[40]	Lightweight communication efficient CNN	Covid-19, Pneumonia, Lung Opacity and Normal X-Ray images classification.	Aggregated dataset from open sources.	3	100	92.96% accuracy
[41]	CNN and Image augmentation	Classification of Covid-19, Pneumonia, Lung Opacity and Normal X-Ray images.	Aggregated dataset from open sources.	3	100	89.43% accuracy in highly non-IID FL setting.
[42]	Federated Blockchain and Capsule Network based FL	Covid-19, Viral Pneumonia and Normal CT images classification.	Open source, GitHub (**https://github.com/abdkhanstd/COVID-19**)	3	600	Capsule network achieved high performance (0.967 Sensitivity)
[48]	COVID-Net MobileNet v2 ResNeXt and ResNet18	Detection of COVID-19 using Pneumonia CXR images.	Open source, (**https://github.com/ieee8023/COVID-chestxray-dataset**)	5	100	ResNet18 performed best with 91.26% sensitivity.
[49]	FedFocus: Integrated CNN and FL	COVID-19 Detection on Chest X-Ray Images	Open source, https://github.com/lindawangg/COVID-Net	10	500	92% accuracy
[50]	Sequential deep learning model-based FL	Analysing COVID19, PNEUMONIA, and NORMAL images and descriptive data for COVID-19 detection	https://www.kaggle.com/datasets/sudalairajkumar/novel-corona-virus-2019-dataseT, https://www.kaggle.com/datasets/prashant268/chest-xray-covid19-pneumonia Open source.	-	10, 50, 200, 500	89.31% accuracy
[52]	GhostNet, ResNet50, and ResNet101	Covid-19, Viral Pneumonia and Negative CT scan or X-ray, to detect COVID-19.	(https://github.com/UCSD-AI4H/COVID-CT), (https://www.kaggle.com/tawsifurrahman/covid19-radiography-database), Open source, Github	3	30	25-30 min less training time and 3200-4200s reduction on upload time.
[51]	An attention mechanism Vision Transformer (ViT) based FL.	Classify COVID-19 and normal patients	Open source. The SIIM-COVID-19 dataset and the normal CXR images from RSNA dataset for client 1. The BIMCV COVID+ dataset and the normal CXR images from the NIH dataset for client 2.	2	100	An Area Under Curve (AUC) of 0.92 and 0.99 for hospital-1 and hospital-2, respectively.
[53]	DMFL_Net + DenseNet-169 model	COVID-19, lung cancer, TB, pneumothorax, and Pneumonia detection	Datasets from several sources such as, The SIRM DB, the TCIA DB, radiopaedia.org, Mendeley, and GitHub	3	100	98.45% accuracy
Our	PFL: ResNet50, VGG-16, InceptionV3	Covid-19, Lung Opacity and Normal X-Ray images classification.	The COVID-19 radiography database (https://www.kaggle.com/tawsifurrahman/covid19-radiography-database)	10, 20	20, 50, 70	95% accuracy for 10C and 10R in just 50 epochs and achieved 91% accuracy in highly non-IID FL setting.

1) 5 epochs are too few to train models. Likewise, a large number of epochs achieves a negligible increase in performance and may increase the model cost. Model tuning is strongly suggested.
2) DL models are developed enough to handle a larger amount of data. Hence, the bigger the data, the better the DL model performance. However, for FL, more enhanced implementations are required to ensure data privacy.
3) With an increased number of clients in FL, more learning rounds can produce notable results.
4) There is a trade-off between accuracy and privacy. We must compromise on accuracy to preserve data privacy.
5) To improve the accuracy of the FedAug implementation for non-IID, it is recommended that a larger number of epochs are used to better train the model.

More enhanced implementations are required to achieve better FL performance. Our experiment results may help other researchers to implement FL.

We may use other metrics to compare model performance, such as Area Under the Curve of the Receiver Operating Characteristic curve AUC ROC in our future work on differentially private FL.

In this work, our main aim was to highlight the importance of FL as a privacy-preserving technique. We did not discuss the communication cost of the experiments under various scenarios. In the future, we may address the communication cost issue and propose a method to ensure privacy under various data scenarios which is a key issue in FL implementation.

Source code: The source code for centralized and Federated learning implementations are available at https://github.com/SadafNaz1980/FL-via-ResNet50/tree/main

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